Phase 1 Report: Association Between Body Mass Index and Systolic Blood Pressure in US Adults

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Abstract

This study investigates the relationship between Body Mass Index (BMI) and Systolic Blood Pressure (SBP) in US adults using data from the National Health and Nutrition Examination Survey (NHANES) 2021-2023. After merging demographic, anthropometric, and blood pressure datasets, a final analytical sample of 6,061 adults aged 18 and older was obtained. Descriptive analyses revealed a slightly right-skewed distribution for BMI and a relatively symmetric distribution for SBP. Initial observations suggest a mild positive association between BMI and SBP, which necessitates further inferential testing in subsequent phases. This report details the data processing pipeline, exploratory data analysis, and essential descriptive statistics.

1 Introduction

Systolic blood pressure is a critical indicator of cardiovascular health, influenced by numerous factors. One such factor is body weight status, commonly measured by Body Mass Index (BMI). The primary objective of this project is to answer the following research question: **Is BMI associated with systolic blood pressure in US adults, even after controlling for confounding variables such as age and gender?** This report presents the initial exploratory and descriptive findings of this investigation.

2 Methods

2.1 Data Source and Study Design

The data for this study were extracted from the National Health and Nutrition Examination Survey (NHANES), a program conducted by the Centers for Disease Control and Prevention (CDC). This is an **observational study**; therefore, causal relationships cannot be established. The data represent the 2021-2023 cycle.

2.2 Data Processing and Analytical Population

Three distinct datasets were merged using the unique identifier `SEQN`: Demographic (`DEMO_L`), Anthropometric (`BMX_L`), and Blood Pressure (`BPXO_L`). The data processing pipeline, including merging and filtering, is summarized in Table 1.

Table 1: Data Flow Summary

Processing Step	**Number of Individuals**
Raw DEMO Data	11,933
After Merging DEMO + BMX	8,860
After Merging with BPXO	7,801
After Dropping Missing Values	7,453
Final Analytical Population (Age ≥ 18)	**6,061**

2.3 Variables

The primary variables of interest are:

- **Response Variable:** Systolic Blood Pressure (SBP), defined as the mean of the three available measurements (`BPXOSY1-3`).
- **Main Explanatory Variable:** Body Mass Index (BMI), derived from anthropometric data (`BMXBMI`).
- **Covariates:** Age (`RIDAGEYR`) and Gender (`RIAGENDR`).
- **Survey Weight:** `WTMEC2YR` was used for weighted analyses.

3 Results

3.1 Descriptive Statistics

The analytical population consisted of 6,061 participants, including 3,313 males (54.66%) and 2,748 females (45.34%). Table 2 presents the summary statistics for the continuous variables.

Table 2: Descriptive Statistics

Variable	**Mean**	**Median**	**SD**	**Variance**	**IQR**
BMI	29.56	28.30	7.19	51.69	8.80
SBP (mmHg)	122.36	120.00	17.91	320.76	21.67

Comparison of the mean and median for BMI indicates a right-skewed distribution, likely due to high-value outliers. Conversely, the distribution of SBP appears relatively symmetric.

3.2 Visualizations

Figures 1 to 4 display the histograms, boxplot, and scatter plot generated to visualize the data distributions and initial associations.

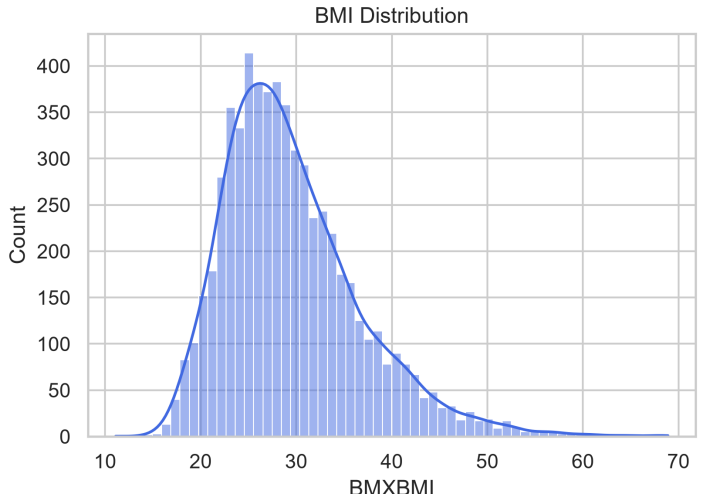


Figure 1: Distribution of Body Mass Index (BMI)

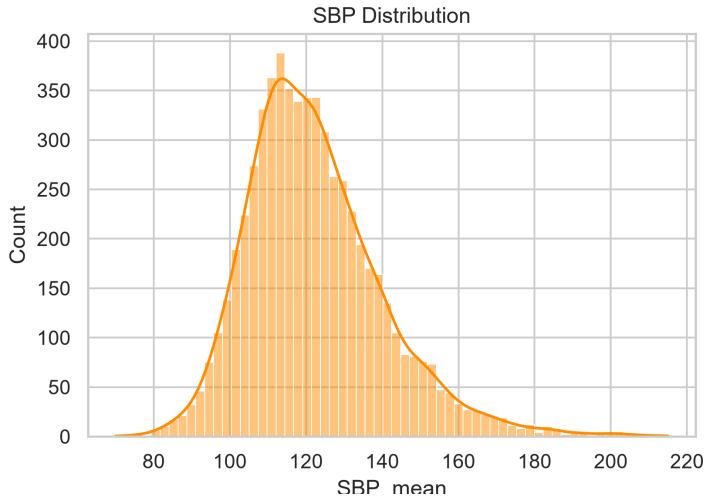


Figure 2: Distribution of Systolic Blood Pressure (SBP)

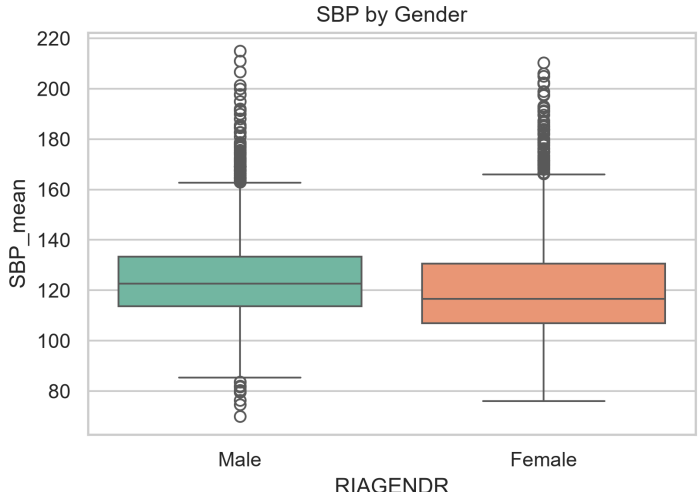


Figure 3: Systolic Blood Pressure by Gender

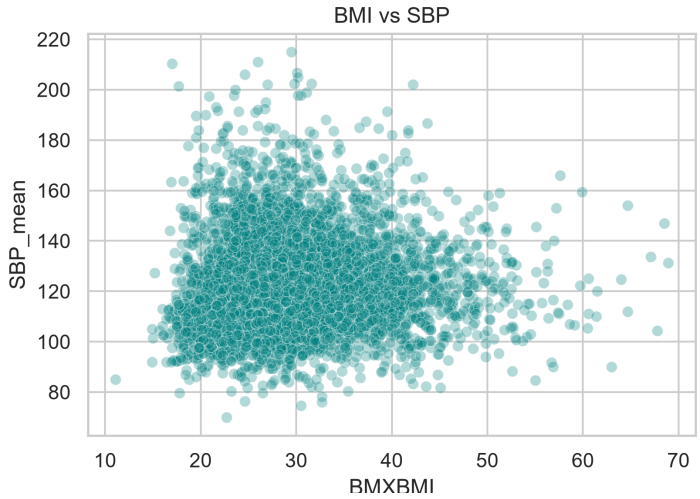


Figure 4: Scatter Plot of BMI vs. Systolic Blood Pressure

3.3 Weighted Analysis

To account for the complex survey design of NHANES, weighted means were calculated. The raw BMI mean was 29.56 (weighted: 29.47), and the raw SBP mean was 122.36 (weighted: 120.77). The discrepancy indicates that individuals with higher blood pressure were slightly underrepresented in the raw sample.

4 Discussion and Conclusion

The data preparation and descriptive analysis confirmed that the merged NHANES dataset is suitable for investigating the primary research question. Initial visual and numerical evidence suggests a potential association between BMI and systolic blood pressure, with gender acting as a potential confounder. In the subsequent phases of this project, formal statistical inference (hypothesis testing and regression modeling) will

be conducted to rigorously test this association while controlling for covariates such as age and gender.

References

- Centers for Disease Control and Prevention (CDC). (2024). National Health and Nutrition Examination Survey (NHANES) 2021-2023. Available at: `https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?Cycle=2021-2023`